Simplex Method

Simplex Method :

① Maximize $Z = 10x_1 + 20x_2$

Subjected to $5x_1 + 3x_2 \leq 30$

$3x_1 + 6x_2 \leq 36$

$2x_1 + 5x_2 \leq 20$

$x_1, x_2 \geq 0$

In Equalities → Equalities.

$\leq$ → Slack

$\geq$ → Surplus

→ $5x_1 + 3x_2 + S_1 = 30$

$3x_1 + 6x_2 + S_2 = 36$

$2x_1 + 5x_2 + S_3 = 20$

$Z - 10x_1 - 20x_2 + 0S_1 + 0S_2 + 0S_3 = 0$

Basic	x_1	x_2	S_1	S_2	S_3	Z	Solution	Ratio
S_1	5	3	1	0	0	0	30	$30/3 = 10$
S_2	3	6 (Key Element)	0	1	0	0	36	$36/6 = 6$
S_3	2	[5]	0	0	1	0	20	$20/5 = 4$ ← Key row
Z	-10	-20	0	0	0	1	0	

↑ Key Column

* Make Key element '1' & all other elements in key column should be made as 'Zero'

New Row III

Operations:

Row I − 2 × New Row III

Row II − 6 × New Row III

Z row − 80 × New Row III + Z row

Basic	x_1	x_2	s_1	s_2	s_3	Z	Soln.	Ratio
s_1	19/5	0	1	0	−3/5	0	18	90/19 ←
s_2	3/5	0	0	1	−6/5	0	12	20
x_2	2/5	1	0	0	1/5	0	4	10
Z	−2	0	0	0	4	+1	80	

New Row I

x_1	1	0	5/19	0	−3/19	0	90/19

Operations.

Row II − 3/5 New R. I

Row III − 2/5 New R I

2 New R I + Z row.

Basic	x_1	x_2	s_1	s_2	s_3	Z	Soln.
x_1	1	0	5/19	0	-3/19	0	90/19
s_2	0	0	-3/19	1	-15	0	136/19
x_2	0	1	-2/19	0	5/19	0	40/19
Z	0	0	10/19	0	20/19	1	1700/19

All the values of Z rows are either 0 or +ve values

optimal solution.

$x_1 = 90/19$

$x_2 = 40/19$

$Z_{max} = 1700/19$

The Big M Method

It is a modified version of the simplex method; in which we assign a very large value (M) to each of the artificial variables.

Problem: Solve the given Linear Programming Problem by Big M Method:
Maximize $z = x_1 + 5x_2$
subject to
$3x_1 + 4x_2 \leq 6$
$x_1 + 3x_2 \geq 2$
$x_1, x_2 \geq 0$

Solution:
Converting inequalities to equalities: By introducing surplus variables, slack variables and artificial variables, the standard form of LPP becomes
Maximize $x_1 + 5x_2 + 0x_3 + 0x_4 – MA_1$
subject to
$3x_1 + 4x_2 + x_3 = 6$
$x_1 + 3x_2 – x_4 + A_1 = 2$
$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, A_1 \geq 0$
where
x_3 is a slack variable
x_4 is a surplus variable.
A_1 is an artificial variable.
Initial basic feasible solution
$x_1 = x_2 = x_4 = 0$
$A_1 = 2, x_3 = 6$

Table 1

	c_j	1	5	0	0	-M	
c_B	Basic Variable B	x_1	x_2	x_3	x_4	A_1	Solution Values $b=(x_B)$
0	x_3	3	4	1	0	0	6
-M	A_1	1	3	0	-1	1	2
$z_j - c_j$		-M-1	-3M-5	0	M	0	

Workings: Calculating values for index row $(z_j – c_j)$
$z_1 – c_1 = 0X_3 + (–M)x_1 – 1 = –M – 1$
$z_2 – c_2 = 0X_4 + (–M)x_3 – 5 = –3M – 5$
$z_3 – c_3 = 0X_1 + (–M)x_0 – 0 = 0$
$z_4 – c_4 = 0X_0 + (–M)x_{(–1)} – 0 = M$
$z_5 – c_5 = 0X_0 + (–M)x_1 – (–M) = 0$
As M is a large positive number, the coefficient of M in the $z_j – c_j$ row would decide the incoming basic variable. As -3M < -M, x_2 becomes a basic variable in the next iteration.
Key column = x_2 column.
Minimum (6/4, 2/3) = 2/3
Key row = A_1 row
Pivot element = 3.
A_1 departs and x_2 enters.

Note that in the iteration just completed, artificial variable A_1 was eliminated from the basis. The new solution is shown in the following table.

Table 2

	c_j	1	5	0	0	
c_B	Basic Variable B	x_1	x_2	x_3	x_4	Solution Values $b=(x_B)$
0	x_3	5/3	0	1	4/3	10/3
5	x_2	1/3	1	0	-1/3	2/3
$z_j - c_j$		2/3	0	0	-5/3	

Table 3

	c_j	1	5	0	0	
c_B	Basic Variable B	x_1	x_2	x_3	x_4	Solution Values $b=(x_B)$
0	x_4	5/4	0	3/4	1	5/2
5	x_2	3/4	1	1/4	0	3/2
$z_j - c_j$		11/4	0	5/4	0	

The optimal solution is:

$x_1 = 0, x_2 = 3/2$

$z = 0 + 5 \text{ X } 3/2 = 15/2$

Problems for Solving:

1. Solve the given Linear Programming Problem by Big M Method:
   Minimize $Z = 40x_1 + 24x_2$
   Subject to
   $20x_1 + 50x_2 \geq 4800$
   $80x_1 + 50x_2 + 6x_3 \geq 7200$
   $x_1, x_2 \geq 0$

2. Solve the given Linear Programming Problem by Big M Method:
   Maximize $Z = 2x_1 + 4x_2$
   Subject to
   $2x_1 + x_2 \leq 18$
   $3x_1 + 2x_2 \geq 30$
   $x_1 + 2x_2 = 26$
   $x_1, x_2 \geq 0$

The Big M Method

It is a modified version of the simplex method; in which we assign a very large value (M) to each of the artificial variables.

Problem: Solve the given Linear Programming Problem by Big M Method:
Maximize $z = x_1 + 5x_2$
subject to

$3x_1 + 4x_2 \leq 6$
$x_1 + 3x_2 \geq 2$
$x_1, x_2 \geq 0$

Solution:
Converting inequalities to equalities: By introducing surplus variables, slack variables and artificial variables, the standard form of LPP becomes
Maximize $x_1 + 5x_2 + 0x_3 + 0x_4 – MA_1$
subject to

$3x_1 + 4x_2 + x_3 = 6$
$x_1 + 3x_2 – x_4 + A_1 = 2$
$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, A_1 \geq 0$

where

x_3 is a slack variable
x_4 is a surplus variable.
A_1 is an artificial variable.

Initial basic feasible solution

$x_1 = x_2 = x_4 = 0$
$A_1 = 2, x_3 = 6$

Table 1

	c_j	1	5	0	0	-M	
c_B	Basic Variable B	x_1	x_2	x_3	x_4	A_1	Solution Values $b=(x_B)$
0	x_3	3	4	1	0	0	6
-M	A_1	1	3	0	-1	1	2
$z_j - c_j$		-M-1	-3M-5	0	M	0	

Workings: Calculating values for index row ($z_j – c_j$)

$z_1 – c_1 = 0X_3 + (–M)x_1 – 1 = –M – 1$
$z_2 – c_2 = 0X_4 + (–M)x_3 – 5 = –3M – 5$
$z_3 – c_3 = 0X_1 + (–M)x_0 – 0 = 0$
$z_4 – c_4 = 0X_0 + (–M)x_{(–1)} – 0 = M$
$z_5 – c_5 = 0X_0 + (–M)x_1 – (–M) = 0$

As M is a large positive number, the coefficient of M in the $z_j – c_j$ row would decide the incoming basic variable. As -3M < -M, x_2 becomes a basic variable in the next iteration.

Key column = x_2 column.
Minimum (6/4, 2/3) = 2/3
Key row = A_1 row
Pivot element = 3.
A_1 departs and x_2 enters.

Note that in the iteration just completed, artificial variable A_1 was eliminated from the basis. The new solution is shown in the following table.

Table 2

	c_j	1	5	0	0	
c_B	Basic Variable B	x_1	x_2	x_3	x_4	Solution Values $b=(x_B)$
0	x_3	5/3	0	1	4/3	10/3
5	x_2	1/3	1	0	-1/3	2/3
$z_j - c_j$		2/3	0	0	-5/3	

Table 3

	c_j	1	5	0	0	
c_B	Basic Variable B	x_1	x_2	x_3	x_4	Solution Values $b=(x_B)$
0	x_4	5/4	0	3/4	1	5/2
5	x_2	3/4	1	1/4	0	3/2
$z_j - c_j$		11/4	0	5/4	0	

The optimal solution is:

$x_1 = 0, x_2 = 3/2$

$z = 0 + 5 \text{ X } 3/2 = 15/2$

Problems for Solving:

1. Solve the given Linear Programming Problem by Big M Method:
   Minimize $Z = 40x_1 + 24x_2$
   Subject to
   $20x_1 + 50x_2 \geq 4800$
   $80x_1 + 50x_2 + 6x_3 \geq 7200$
   $x_1, x_2 \geq 0$

2. Solve the given Linear Programming Problem by Big M Method:
   Maximize $Z = 2x_1 + 4x_2$
   Subject to
   $2x_1 + x_2 \leq 18$
   $3x_1 + 2x_2 \geq 30$
   $x_1 + 2x_2 = 26$
   $x_1, x_2 \geq 0$

* Two Phase Simplex Method *

LPP can be solved by Two phase simplex method provided it contains ≥ or = type constraints where Artificial Variable is required. As the name suggest, it separates the solution procedure into Two Phases.

Phase 1:

1. Express the given LPP in the standard form.
2. Convert each of the constraints into equality by introducing slack or surplus or artificial variables.
3. Assign a cost +1 to each artificial variable in case of minimisation & −1 in case of maximisation and a cost 0 to all other variables irrespective of the given cost and formulate a new objective function Z
4. The arrangement of the stated co-efficients yields the following auxiliary LPP

$$\text{Min } Z = \sum_{j=1}^{n} 0x_j + \sum_{i=1}^{m} 0S_i + \sum_{i=1}^{m} 1A_i$$

Subject to

$$\sum_{j=1}^{n} a_{ij} x_j - S_i + A_i = b_i \qquad i = 1, 2, 3, \ldots m$$

and $x_j, S_i, A_i \geq 0$ for all $i \& j$

5. Solve the auxiliary LPP by simplex method & obtain the optimum basic feasible solution.

If $c_j - Z_j$ row indicates optimal solution (as it contains all 0 or −ve elements) in case of max. problem and 0 or +ve in case of minimisation if:

(a) the Artificial variable appears as a basic variable. The given LPP has infeasible soln., we terminate our soln in Phase I.

(b) The Artificial variable does not appear as a basic variable. The given LPP has a feasible soln. and we proceed to Phase II.

Phase II:

i) Assign the actual cost c_j to the variables in the objective function and start with the (optimal) soln ~~obtained~~ contained in the final simplex tableau of the Phase I.

ii) Apply simplex algorithm to the problem contained in the new table to obtain the optimal solution.

iii) The optimum basic feasible solution thus obtained will be then optimum basic feasible solution to the original LPP.

Example:

I. Minimisation when all constraints are ≥ 0:

1. Use Two-Phase Simplex Method to solve LPP:

Min $Z = \frac{15}{2} x_1 - 3x_2$

Subject to

$$3x_1 - x_2 - x_3 \geq 3$$
$$x_1 - x_2 + x_3 \geq 2$$
$$x_1, x_2, x_3 \geq 0$$

Solution:

Std. Form:

Min $Z = 0x_1 + 0x_2 + 0x_3 + 0S_1 + 0S_2 + 1A_1 + 1A_2$

Subject to $3x_1 - x_2 - x_3 - S_1 + 0S_2 + A_1 + 0A_2 = 3$

$x_1 - x_2 + x_3 + 0S_1 - S_2 + 0A_1 + A_2 = 2$

$x_1, x_2, x_3, S_1, S_2, A_1, A_2 \geq 0$

$C_j \rightarrow$	Basic Variable	Sol.?	0 x_1	0 x_2	0 x_3	0 S_1	0 S_2	1 A_1	1 A_2	Min. Ratio
1	A_1	3	[3] ← Key	-1	-1	-1	0	1	0	3/3 = 1 ← Min.
1	A_2	2	1	-1	1	0	-1	0	1	2/1 = 2
	Z_j	5	4	-2	0	-1	-1	1	1	
	$C_j - Z_j$		-4	2	0	1	1	0	0	

(3×1 + 2×1) ↑ Key column [-1×1 + (-1)×1]

Operations:

1. Row I ÷ 3
2. Row II − New Row I
3. Z row − 4 New Row I.

$C_j \rightarrow$	Basic Variable	Sol.?	0 x_1	0 x_2	0 x_3	0 S_1	0 S_2	1 A_1	1 A_2	Min. Ratio
0	x_1	1	1	-1/3	-1/3	-1/3	0	-	0	-ve
1	A_2	1	0	-2/3	[4/3] ← Key	1/3	-1	-	1	3/4 ← Min.
	Z_j	1	0	-2/3	4/3	1/3	-1	-	1	
	$C_j - Z_j$		0	2/3	-4/3	-1/3	1	-	0	

Operations:

1. New Row II = Row II ÷ 4/3
2.
3.

$C_j \rightarrow$	Basic Variable	Sol.?	0 x_1	0 x_2	0 x_3	0 S_1	0 S_2
0	x_1	5/4	1	-1/2	0	-1/4	-1/4
0	x_3	3/4	0	-1/2	1	1/4	-3/4
	Z_j	0	0	0	0	0	0
	$C_j - Z_j$		0	0	0	0	0

Since $C_j - Z_j \geq 0$ and no artificial variables appears in the basis. Therefore an optimum basic feasible sol.? of the auxiliary LPP has been

obtained. Therefore, we proceed to Phase II.

Phase II:

The objective function

$$\text{Min } Z = \frac{15}{2}x_1 - 3x_2 + 0x_3 + 0s_1 + 0s_2$$

Now we replace c_j value in the optimum table of Phase I & take it as starting simplex table of Phase II.

Phase II:

$c_j \rightarrow$			$15/2$	-3	0	0	0
	Basic Variable	Sol?.	x_1	x_2	x_3	s_1	s_2
$15/2$	x_1	$5/4$	1	$-1/2$	0	$-1/4$	$-1/4$
0	x_3	$3/4$	0	$-1/2$	1	$1/4$	$-3/4$
	Z_j	$75/8$	$15/2$	$-15/4$	0	$-15/8$	$-15/8$
	$c_j - Z_j$		0	$3/4$	0	$+15/8$	$+15/8$

$(15/2 \times 1 + 0 \times 0) = 15/2$

$(15/2 \times 5/4 + 0 \times 3/4) = 75/8$

Since all elements in $c_j - Z_j \geq 0$, the optimum solution is

$$x_1 = 5/4,\ x_2 = 0,\ x_3 = 3/4,\ Z = 75/8$$

II. Maximisation case when all constraints are equal:

Example: Solve the following LP problem by Two Phase simplex Method:

$$\text{Max } Z = x_1 + 2x_2 + 3x_3 - x_4$$

Subjected to $x_1 + 2x_2 + 3x_3 = 15$

$$2x_1 + x_2 + 5x_3 = 20$$

$$x_1 + 2x_2 + x_3 + x_4 = 10$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Solution: Introducing artificial variables & constraints
Phase I : & assigning zero to basic variables & -1 coeff cost to artificial variables

Max $Z = 0x_1 + 0x_2 + 0x_3 - 0x_4 - A_1 - A_2 - A_3$

Subject to

$x_1 + 2x_2 + 3x_3 + A_1 + 0A_2 + 0A_3 = 15$

$2x_1 + x_2 + 5x_3 + 0A_1 + A_2 + 0A_3 = 20$

$x_1 + 2x_2 + x_3 + x_4 + 0A_1 + 0A_2 + A_3 = 10$

$x_1, x_2, x_3, A_1, A_2, A_3 \geq 0$.

$C_j \rightarrow$			0	0	0	0	-1	-1	-1	
↓	Basic Variable	Sol^n	x_1	x_2	x_3	x_4	A_1	A_2	A_3	Min Ratio
-1	A_1	15	1	2	3	0	1	0	0	5
-1	A_2	20	2	1	[5]	0	0	1	0	4 →
-1	A_3	10	1	2	1	1	0	0	1	10
	Z_j		-4	-5	-9	-1	-1	-1	-1	
	$C_j - Z_j$		4	5	9 ↑	1	0	0	0	

Operations.

1. New Row II = Row II ÷ 5
2. Row I − 3 New Row II
3. Row III − New Row II
4. Z row + 9 New Row II.

$C_j \rightarrow$			0	0	0	0	-1	-1	-1	
↓	Basic Variable	Sol^n	x_1	x_2	x_3	x_4	A_1	A_2	A_3	Min Ratio
-1	A_1	3	-1/5	[7/5]	0	0	1	-	0	15/7 →
0	x_3	4	2/5	1/5	1	0	0	-	0	20
-1	A_3	6	3/5	9/5	0	1	0	-	1	30/9
	Z_j		-2/5	-16/5	0	-1	-1	-	-1	
	$C_j - Z_j$		2/5	16/5 ↑	0	1	0	-	0	

Operations:

1. New Row I = Row I ÷ 7/5
2.

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$C_j \rightarrow$			0	0	0	0	−1	−1	−1	
↓	Basic Variable	Sol?. Value	x_1	x_2	x_3	x_4	A_1	A_2	A_3	Min Ratio
0	x_2	15/7	−1/7	1	0	0	−	−	0	∞
0	x_3	25/7	3/7	0	1	0	−	−	0	∞
−1	A_3	15/7	6/7	0	0	[1]	−	−	1	5/7 →
	Z_j	−15/7	−6/7	0	0	−1	−	−	−1	
	$C_j - Z_j$		6/7	0	0	1	−	−	0	

↑

Operations:

1.

2.

3.

$C_j \rightarrow$			0	0	0	0
↓	Basic Variable	Sol?.	x_1	x_2	x_3	x_4
0	x_2	15/7	−1/7	1	0	0
0	x_3	25/7	3/7	0	1	0
0	x_4	15/7	6/7	0	0	1
	Z_j	0	0	0	0	0
	$C_j - Z_j$		0	0	0	0

Since $C_j - Z_j$ are zero & no artificial variable appears as basic variable we proceed to Phase II.

Phase II: Max $Z = x_1 + 2x_2 + 3x_3 - x_4$

$C_j \rightarrow$			1	2	3	−1	
↓	Basic Variable	Sol?.	x_1	x_2	x_3	x_4	Min Ratio
2	x_2	15/7	−1/7	1	0	0	−15
3	x_3	25/7	3/7	0	1	0	25/3
−1	x_4	15/7	[6/7]	0	0	1	5/2 →
	Z_j	90/7	4/7	2	3	−1	
	$C_j - Z_j$		6/7	0	0	0	

↑

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Operations:

1.
2.
3.

$C_j \rightarrow$			1	2	3	-1
↓	Basic Variable	Sol?	x_1	x_2	x_3	x_4
2	x_2	5/2	0	1	0	1/6
3	x_3	5/2	0	0	1	-1/2
1	x_1	5/2	1	0	0	7/6
	Z_j	15	0	2	3	0
	$C_j - Z_j$		0	0	0	-1

Since there is no positive values, the optimal solution is

$x_1 = 5/2$ $\quad x_3 = 5/2$ $\quad Z_{max} = 15$

$x_2 = 5/2$ $\quad x_4 = 0$

III. Minimisation case with mixed constraints:

Example: An animal feed company must produce 200 kg of a mixture consisting of ingredient x_1 & x_2 daily. x_1 cost Rs 3 per kg and x_2 Rs. 8 per kg. No more than 80 kg of x_1 can be used and at least 60 kg of x_2 must be used. Find how much of each ingredient should be used if the company wants to minimise cost, use Two Phase Method.

Solution: LPP formulation

Minimise $\quad Z = 3x_1 + 8x_2$ $\quad$ Cost function

Subject to $\quad x_1 + x_2 = 200$

$x_1 \leq 80$ } Mixture Quantity Constraint

$x_2 \geq 60$ }

$x_1, x_2 \geq 0$

PRESTIGE

Standard Form:

Min $Z = 0x_1 + 0x_2 + 0S_1 + 0S_2 + A_1 + A_2$

Subject to $x_1 + x_2 + A_1 = 200$

$x_1 + 0S_2 + S_1 = 80$

$0x_1 + x_2 - S_2 + A_2 = 60$

where $x_1, x_2, S_1, S_2, A_1, A_2$ all ≥ 0

Phase I:

$C_j \rightarrow$			0	0	0	0	1	1	
↓	Basic Variable	Sol? Values	x_1	x_2	S_1	S_2	A_1	A_2	Min Ratio
1	A_1	200	1	1	0	0	1	0	200
0	S_1	80	1	0	1	0	0	0	∞
1	A_2	60	0	[1]	0	−1	0	1	60 →
	Z_j	260	1	2	0	−1	1	1	
	$C_j - Z_j$		−1	−2 ↑	0	1	0	0	

Operations:

1.

2.

3.

4.

$C_j \rightarrow$			0	0	0	0	1	1	
↓	Basic Variable	Sol? Values	x_1	x_2	S_1	S_2	A_1	A_2	Min Ratio
1	A_1	140	1	0	0	1	1	-	140
0	S_1	80	[1]	0	1	0	0	-	80 →
0	x_2	60	0	1	0	−1	0	-	∞
	Z_j		1	0	0	1	1	-	
	$C_j - Z_j$		−1 ↑	0	0	−1	0	-	

Operations

$C_j \rightarrow$			0	0	0	0	1	1	
↓	Basic Variable	Values	x_1	x_2	S_1	S_2	A_1	A_2	Min Ratio
1	A_1	60	0	0	−1	[1]	1	−	60 →
0	x_1	80	1	0	1	0	0	−	80
0	x_2	60	0	1	0	−1	0	−	60
	Z_j		0	0	−1	1	1	−	
	$C_j - Z_j$		0	0	1	−1 ↑	0	−	

Operations:

1.
2.
3.
4.

$C_j \rightarrow$			0	0	0	0
	Basic Variable	Values	x_1	x_2	S_1	S_2
0	S_2	60	0	0	−1	1
0	x_1	80	1	0	1	0
0	x_2	120	0	1	−1	0
	Z_j		0	0	0	0
	$C_j - Z_j$		0	0	0	0

Since all $C_j - Z_j \geq 0$, we move to Phase II

Phase II:

Min $Z = 3x_1 + 8x_2 + 0S_1 + 0S_2$

$C_j \rightarrow$			3	8	0	0
↓	Basic Variable	Values	x_1	x_2	S_1	S_2
0	S_2	60	0	0	−1	1
3	x_1	80	1	0	1	0
8	x_2	120	0	1	−1	0
	Z_j	1200	3	8	−5	0
	$C_j - Z_j$		0	0	5	0

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since $c_j - z_j$ has no -ve value, the optimal solution is

Min $Z = 1200$ $x_1 = 80$ $x_2 = 120$

IV. Maximisation case with mixed constraints.

Example: A manufacture produces two different models x & y of the same product. The raw materials r_1 & r_2 are required for production. At least 18 kg of r_1 and 12 kgs of r_2 must be used daily. Also at most 34 hours of labour are to be utilised; 2 kg of r_1 are needed for each model x and 1 kg of r_1 for each model y. For each model x & y, 1 kg of r_2 is required. It takes 3 hours to manufacture a model x and 2 hours to manufacture a model y. The profit is Rs. 50 for each model x and Rs. 30 for each model y. How many of units of each model should be produced to maximise the profit. Use two phase simplex method.

Sol^n: LPP formulation

Max $Z = 50x_1 + 30x_2$ Profit function

Subject to $2x_1 + x_2 \geq 18$ } Quantity constraint

$x_1 + x_2 \geq 12$ }

$3x_1 + 2x_2 \leq 34$ Labour hr. constraint

$x_1, x_2 \geq 0$

Phase I:

Max $Z = 0x_1 + 0x_2 + 0S_1 + 0S_2 + 0S_3 - A_1 - A_2$

Subject to $2x_1 + x_2 - S_1 + A_1 = 18$

$x_1 + x_2 - S_2 + A_2 = 12$

$3x_1 + 2x_2 + S_3 = 34$

$x_1, x_2, S_1, S_2, S_3, A_1, A_2$ all ≥ 0

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$C_j \rightarrow$			0	0	0	0	0	-1	-1	
↓	Basic Variable	Sol? Values	x_1	x_2	S_1	S_2	S_3	A_1	A_2	Min Ratio
-1	A_1	18	[2]	1	-1	0	0	1	0	9 →
-1	A_2	12	1	1	0	-1	0	0	1	12
0	S_3	34	3	2	0	0	1	0	0	34/3
	Z_j	-30	-3	-2	1	1	0	-1	-1	
	$C_j - Z_j$		3 ↑	2	-1	-1	0	0	0	

Operations:

1.
2.
3.
4.

$C_j \rightarrow$			0	0	0	0	0	-1	
↓	Basic Variable	Sol? Values	x_1	x_2	S_1	S_2	S_3	A_2	Min Ratio
0	x_1	9	1	1/2	-1/2	0	0	0	-18
-1	A_2	3	0	1/2	1/2	-1	0	1	6
0	S_3	7	0	1/2	[3/2]	0	1	0	14/3 →
	Z_j	-3	0	-1/2	-1/2	1	0	-1	
	$C_j - Z_j$		0	1/2	1/2 ↑	-1	0	0	

Operations.

1.
2.
3.
4.

$C_j \rightarrow$			0	0	0	0	0	-1	
↓	Basic Variable	Sol? Values	x_1	x_2	S_1	S_2	S_3	A_2	Min Ratio
0	x_1	34/3	1	2/3	0	0	1/3	0	17
-1	A_2	2/3	0	[1/3]	0	-1	-4/3	1	2 →
0	S_1	14/3	0	1/3	1	0	2/3	0	14
	Z_j	-2/3	0	-1/3	0	1	1/3	-1	
	$C_j - Z_j$		0	1/3 ↑	0	1	-1/3	0	

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Operations:

1.

2.

3.

4.

$C_j \rightarrow$			0	0	0	0	0
$\downarrow$	Basic Variable	Sol^n Value	x_1	x_2	s_1	s_2	s_3
0	x_1	10	1	0	0	2	1
0	x_2	2	0	1	0	-3	-1
0	s_1	4	0	0	1	1	1
	Z_j	0	0	0	0	0	0
	$C_j - Z_j$		0	0	0	0	0

Since $C_j - Z_j$ are 0, we move to Phase II.

Phase II:

Max $Z = 50x_1 + 30x_2 + 0s_1 + 0s_2 + 0s_3$

$C_j \rightarrow$			50	30	0	0	0
$\downarrow$	Basic Variable	Sol^n Value	x_1	x_2	s_1	s_2	s_3
50	x_1	10	1	0	0	2	1
30	x_2	2	0	1	0	-3	-1
0	s_1	4	0	0	1	1	1
	Z_j	560	50	30	0	10	20
	$C_j - Z_j$		0	0	0	-10	-20

Since all the values of $C_j - Z_j$ are 0 or -ve, the optimal solution

$x_1 = 10$ $\quad Z_{max} = 560$

$x_2 = 2$

8. Minimisation case with equal constraints.

Example:

Min $Z = x_1 - 2x_2 - 3x_3$

Subject to $-2x_1 + x_2 + 3x_3 = 2$

$2x_1 + 3x_2 + 4x_3 = 1$

$x_1, x_2, x_3 \geq 0$

Solution: Phase I.

Min $Z = 0x_1 - 0x_2 - 0x_3 + A_1 + A_2$

Subject to $-2x_1 + x_2 + 3x_3 + A_1 = 2$

$2x_1 + 3x_2 + 4x_3 + A_2 = 1$

x_1, x_2, x_3, A_1, A_2 all ≥ 0

$C_j \rightarrow$			0	0	0	1	1	
$\downarrow$	Basic Variable	Sol? Values	x_1	x_2	x_3	A_1	A_2	Min Ratio
1	A_1	2	-2	1	3	1	0	2/3
1	A_2	1	2	3	[4]	0	1	1/4 →
	Z_j	3	0	4	7	1	0	
	$C_j - Z_j$		0	-4	-7 ↑	0	0	

Operation

1.

2.

3.

4.

$C_j \rightarrow$			0	0	0	1	1
$\downarrow$	Basic Variable	Sol? Values	x_1	x_2	x_3	A_1	A_2
1	A_1	5/4	-7/2	-5/4	0	1	-
0	x_3	1/4	1/2	3/4	1	0	-
	Z_j	5/4	-7/2	-5/4	0	1	-
	$C_j - Z_j$		7/2	5/4	0	0	-

Since all $C_j - Z_j$ are 0 or +ve, but artificial variable is present as basic variable, the problem has got infeasible solution.

The Dual Problem: Minimization with problem constraints of the form ≥

- Linear programming problems exist in **pairs**. That is in linear programming problem, every **maximization** problem is associated with a **minimization** problem. Conversely, associated with every minimization problem is a maximization problem. Once we have a problem with its objective function as maximization, we can write by using duality relationship of linear programming problems, its minimization version. The original linear programming problem is known as **primal problem**, and the derived problem is known as **dual problem**.

To start the dual Simplex method, the following three conditions are to be met:

1. **The objective function must satisfy the optimality conditions of the regular Simplex method.**

2. **All the constraints must be of the type $\leq$.**

3. **All variables should be ≥ 0.**

(**Note**: As in the simplex method, we must have an identity matrix in the constraint matrix; however, the RHS constants b_i need NOT be ≥ 0.)

In any iteration, we first decide the “leaving” variable and then decide which variable should enter (the basis).

Dual Feasibility Condition

(=Condition for a variable to leave (the basis))

The leaving variable, x_i, is the basic variable having the most **negative** value (i.e. in the Simplex tableau, the corresponding constraint row has the most negative RHS). Ties are broken arbitrarily. If all the basic variables are ≥ 0, the algorithm ends and we have obtained the optimal solution.

The dual problem uses exactly the same parameters as the primal problem, but in different locations. To highlight the comparison, now look at these same two problems in matrix notation.

Primal Problem			Dual Problem	
Minimize	Z=cx		Maximize	W=yb
Subject to	Ax≥b		Subject to	yA≤c
and	x≥0		And	y≥0

Primal problem

$$A = \left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ b_{11} & b_{12} & b_{13} \\ \hline c_{11} & c_{12} & c_{13} \end{array}\right]$$

Dual problem

$$A^T = \left[\begin{array}{ccc} a_{11} & b_{11} & c_{11} \\ a_{12} & b_{12} & c_{12} \\ \hline a_{13} & b_{13} & c_{13} \end{array}\right]$$

Example :Solve by Dual Simplex method

Minimize

$$z = 4x_1 + 2x_2$$

subject to

$$x_1 + x_2 = 1$$
$$3x_1 - x_2 \geq 2$$
$$x_1, x_2 \geq 0$$

Note this !



That is subject to

$$x_1 + x_2 \leq 1$$
$$-x_1 - x_2 \leq -1$$
$$-3x_1 + x_2 \leq -2$$
$$x_1, x_2 \geq 0$$

Adding slack variables, the problem becomes

Minimize $z = 4x_1 + 2x_2 + 0s_1 + 0s_2 + 0s_3$

subject to

$$\begin{aligned} x_1 + x_2 + s_1 &= 1 \\ -x_1 - x_2 + s_2 &= -1 \\ -3x_1 + x_2 + s_3 &= -2 \\ x_1, x_2, s_1, s_2, s_3 &\geq 0 \end{aligned}$$

Since all the objective coefficients are ≥ 0 and the problem is minimization, optimality is met. We start the solution by writing the Simplex tableau.

Basic	z	x1	x2	s1	s2	s3	Sol
z	1	-4	-2	0	0	0	0
s1	0	1	1	1	0	0	1
s2	0	-1	-1	0	1	0	-1
s3	0	-3	1	0	0	1	-2
z	1	0	-10/3	0	0	-4/3	8/3
s1	0	0	4/3	1	0	1/3	1/3
s2	0	0	-4/3	0	1	-1/3	-1/3
x1	0	1	-1/3	0	0	-1/3	2/3
z	1	0	0	0	-5/2	-1/2	7/2
s1	0	0	0	1	1	0	0
x2	0	0	1	0	-3/4	1/4	1/4
x1	0	1	0	0	-1/4	-1/4	3/4

Optimal Tableau

Summary

Primal	Dual
(a) Maximize.	Minimize
(b) Objective Function.	Right hand side.
(c) Right hand side.	Objective function.
(d) i th row of input-output coefficients.	i th column of input output coefficients.
(e) j th column of input-output coefficients.	j the row of input-output coefficients.

Solution of Minimization Problems

ORIGINAL PROBLEM (1)	DUAL PROBLEM (2)
Minimize $C = 16x_1 + 45x_2$	**Maximize** $P = 50y_1 + 27y_2$
subject to $2x_1 + 5x_2 \geq 50$ $x_1 + 3x_2 \geq 27$ $x_1, x_2 \geq 0$	subject to $2y_1 + y_2 \leq 16$ $5y_1 + 3y_2 \leq 45$ $y_1, y_2 \geq 0$

CHAPTER-6: SIMPLEX & DUAL SIMPLEX METHOD

1. © **Simplex method**

Introduction

Simplex method = Simplex technique = Simplex algorithm.

It is an iterative procedure for solving a linear programming problem in a finite no. of steps. This method provides an algorithm which consist in moving from one vertex of the region of feasible solution to another in such a way that the value of the objective function at the succeeding vertex is less (or more) than the preceding vertex so as to reach finally in the optimum solution.

The following important definitions are necessary to understand the simplex method.

(a) **Basic solution**

Given a system 'm' simultaneous linear equations in 'n' unknowns (m<n).

Ax = b,

Where A is a m×n matrix of rank m. Let B be any m×m submatrix formed by m linearly in dependent columns of A. Then a solution obtained by setting n – m variable not associated with the columns of B equals to zero and solving the resulting system is called a basic solution to the given system of equations.

This can be explained with the following example.

Ex: Obtain basic solution to $x_1+2x_2+x_3 = 4$ ----------(1)

$2x_1+x_2+5x_3 = 5$ ----------(2)

In this case m = 2 (i.e., Eq. (1) & Eq. (2))

n = 3 (i.e., x_1, x_2, x_3) = m<n

the above system of equation can be written as:

$$\begin{pmatrix}1 & 2 & 1\\ 2 & 1 & 5\end{pmatrix}\begin{bmatrix}x1\\ x2\\ x3\end{bmatrix}=\begin{bmatrix}4\\ 5\end{bmatrix}$$

A x b

$(2\times3)\times(3\times1)=(2\times1)$

The different sub matrix from matrix A are:

$$.\begin{pmatrix}1 & 2\\ 2 & 1\end{pmatrix},\begin{pmatrix}1 & 1\\ 2 & 5\end{pmatrix}\&\begin{pmatrix}2 & 1\\ 1 & 5\end{pmatrix}$$

The basic solutions are obtained from the following equations:

(I) $\begin{pmatrix}1 & 2\\ 2 & 1\end{pmatrix}\begin{bmatrix}x1\\ x2\end{bmatrix}=\begin{bmatrix}4\\ 5\end{bmatrix}$-------(1) setting $x_3 = 0$

(II) $\begin{pmatrix}2 & 1\\ 1 & 5\end{pmatrix}\begin{bmatrix}x2\\ x3\end{bmatrix}=\begin{bmatrix}4\\ 5\end{bmatrix}$ ----------(2) setting $x_1 = 0$

(III) $\begin{pmatrix}1 & 1\\ 2 & 5\end{pmatrix}\begin{bmatrix}x1\\ x2\end{bmatrix}=\begin{bmatrix}4\\ 5\end{bmatrix}$----------(3) setting $x_2 = 0$

Considering Eq (1)

$[x_1+2x_2 = 4]\times2$

$\underline{2x_1+x_2 = 5}$

$4x_2-x_2 = 8 - 5 = 3$

$= 3x_2 = 3 = x_2 = 1$

$2x_1+1 = 5 = x_1 = 2$

So basic $x_1 = 2$, $x_2 = 1$ & non basic $x_3 = 0$

Similarly solving other 2 equations :

Basic $x_2 = 5/3$, $x_3 = 2/3$ non basic $x_1 = 0$

Basic $x_1 = 5$, $x_3 = -1$ non basic $x_2 = 0$

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(b) **Degenerate solution**

A basic solution to the system is called Degenerate of one or more of the basic variables vanish.

Ex: Show that the following system of linear equations has a degenerate solution.

2x1+x2-x3 = 2

3x1+2x2+x3 = 3

Solution

The given system of linear equations can be written as:

$$\cdot\begin{pmatrix}2 & 1 & -1\\ 3 & 2 & 1\end{pmatrix}\begin{bmatrix}x1\\ x2\\ x3\end{bmatrix}=\begin{bmatrix}2\\ 3\end{bmatrix}$$

A x b

The basic solution can be obtained as:

$$\cdot\begin{pmatrix}2 & 1\\ 3 & 2\end{pmatrix}\begin{bmatrix}x1\\ x2\end{bmatrix}=\begin{bmatrix}2\\ 3\end{bmatrix}$$----------(1) by setting $x_3 = 0$

$$\cdot\begin{pmatrix}1 & -1\\ 2 & 1\end{pmatrix}\begin{bmatrix}x2\\ x3\end{bmatrix}=\begin{bmatrix}2\\ 3\end{bmatrix}$$----------(2) by setting $x_1 = 0$

$$\cdot\begin{pmatrix}2 & -1\\ 3 & 1\end{pmatrix}\begin{bmatrix}x1\\ x3\end{bmatrix}=\begin{bmatrix}2\\ 3\end{bmatrix}$$----------(3) by setting $x_2 = 0$

Solving (1) Basic → $x_1 = 1$, $x_2 = 0$; non basic $x_3 = 0$

(2) Basic → $x_2 = 5/3$, $x_3 = -1/3$; non basic $x_1 = 0$

(3) Basic → $x_1 = 1$, $x_3 = 0$; non basic $x_2 = 0$

In each is 2 basic solutions at least one of the basic variables is zero. Hence two of the basic solutions are degenerate solutions.

(c) Basic feasible solutions

A feasible solution to a LPP which is also basic solution to the problem is called as basic feasible solution to LPP. Let the LPP be to determine x so as to maximize

$z = cx$

subjected to constraints:

$Ax = b$

& $x \geq 0$

Then x_B is a basic feasible solution to this problem if B is an m×m non-singular sub-matrix of A and $Bx_B = b$ $x_B \geq 0$

Looking to the solutions of earlier problem.

(1)

$x_2 = \frac{5}{3}, x_3 = -\frac{1}{3}(negative), x_1 = 0 \leftarrow$

$not\ basic\ feasible\ solution\ since\ one\ soln\ is - ve$

(2) x1 = 1, x2 = 0 (basic); x3 = 0 (non basic)

Basic feasible solution

(3) x1 = 1, x3 = 0(basic) ; x2 = 0 (non basic)

(d) Associated cost vector

Let x_B be a basic feasible solution to the LPP:

Maximize z = cx

Subject to constraint Ax = b and x ≥ 0.

The vector $c_B = (c_{B1}, c_{B2}\ldots\ldots\ldots c_{Bm})$ where c_{Bi} are components of c associated with the basic variables is called the cost vector associated with the basic feasible solution x_B.

It is obvious that the value of the objective function for the basic feasible solution x_B, is given by:

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$z_0 = c_B x_B$

(e) Improved Basic Feasible Solution

Let x_B and $\hat{x}_B$ be two basic feasible solution to the standard LPP, then $\hat{x}_B$ is said to be an improved basic feasible solution as compared to x_B if

$$\hat{c}_B \hat{x}_B \geq c_B x_B$$

Where $\hat{c}_B$ is constituted of cost components corresponding to $\hat{x}_B$.

(f) Optimum Basic Feasible Solution

A basic feasible solution xB to LPP –

Maximize z = cx

Subjected to Ax = b and x ≥ 0.

Is called an optimum basic feasible solution if

$z_0 = c_B x_B \geq z^*$

where z* is value of objective function for any feasible solution.

© Fundamental Properties of solution

The fundamental properties of solution will allow us to arrive at the simplex algorithm. These properties are presented in the following in the form of theorems:

Theorem- 1: (Reduction of Feasible solution to basic feasible solution) If a LPP has a feasible solution, then it also has a basic feasible solution.

Theorem-2: (Replacement of a basic vector) Let a LPP have a basic feasible solution. If we drop one of the basic vectors and introduce a non-basic vector in the basic set, then the new solution obtained is also a basic feasible solution.

Theorem- 3: Improved basic feasible solution – stated earlier.

Theorem- 4 : Unbounded solution

Let there exist a basic feasible solution to a given LPP. If for at least one j, for which yij ≤ 0 (i = 1,2,3,……..m), zj –cj is negative, then there does not exist any optimum solution to this LPP.

Theorem-5: condition of optimality

A sufficient condition for a basic feasible solution to a LPP to be an optimum (maximum) is that zj – cj ≥ 0 for all j for which the column vector is not in the basis.

Theorem- 6: Any convex combination of k different optimum solutions to a LPP is again on optimum solution to the problem.

Theorem- 7: Minimax theorem

Let f be a linear function of n variables such that f(x*) is its minimum value for some point x^{T*}, x^T E R^n, Then – f(x) attains its maximum at the point x*. Moreover for x^T E R^n.

Minimum f(x) = -maximum{- f(x)}

© **Computational procedure**

The computational procedure for simplex algorithm can be explained with the help of a typical example.

Use simplex algorithm to solve the following LPP.

Maximize z = 4x1+5x2+9x3

Subjected to constraints : $x_1+x_2+2x_3 \leq 16$

$fx_1+5x_2+3x_3 \leq 25$

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$

Solution:

Step-1: Check whether the objective function is to maximize or minimize. if it is a minimization case, then convert it into maximization by this relation minimize z = maximize (-z).

Step-2: Check whether all bi are +ve. If any one bi is –ve, multiply the corresponding in equation by (-1).

Step-3: Convert all the inequations of constraints into equation by introducing slack/surplus variables

Now introducing slack variable, x_4 & x_5 the inequation of constraints because:

$X_1+x_2+2x_3+x_4 = 16$ ----------(1)

$fx_1+5x_2+3x_3+x_5 = 25$ ----------(2)

and hence the new objected function becomes:

$z = 4x_1+5x_2+9x_3+0.x_4+0.x_5$ ----------(3)

step-4: Obtain an initial basic feasible solution to the problem in the form of

$x_B = {}^{-1}_{B}(b)$ and put in the x_B column of the simplex table.

The set of equation (1) & (2) can be written as:

$$\cdot\begin{pmatrix}1 & 1 & 2 & 1 & 0\\ 7 & 5 & 3 & 0 & 1\end{pmatrix}\begin{bmatrix}x1\\ x2\\ x3\\ x4\\ x5\end{bmatrix} = \begin{bmatrix}16\\ 25\end{bmatrix}$$

To obtain an initial basic feasible solution more easily the following matrix equation can be considered

i.e.

$$\cdot\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{bmatrix}x4\\ x5\end{bmatrix} = \begin{bmatrix}16\\ 25\end{bmatrix}$$

:- x4 = 16

& x5 = 25

These values are substituted in the appropriate place in the initial simplex table.

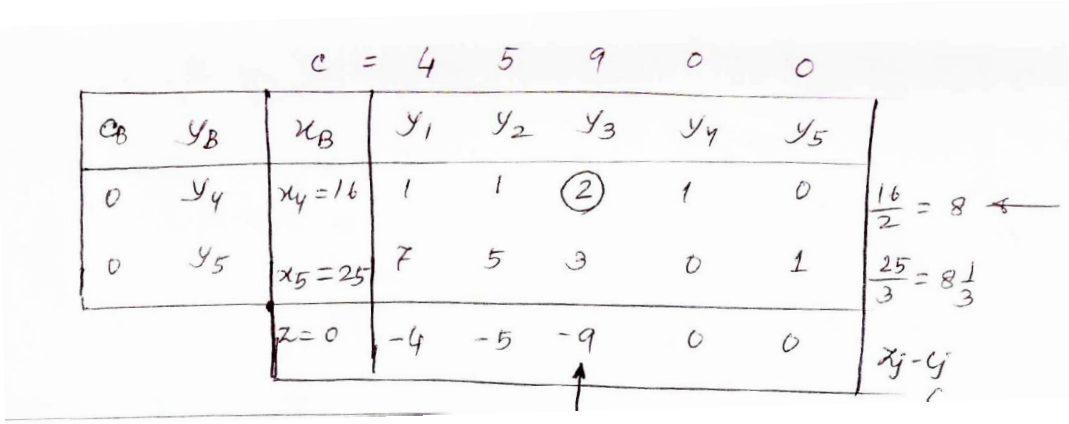
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Step-5: Compute the net evaluation i.e., $z_j - c_j$ (j = 1,2,……..n) by using the relation $z_j - c_j = c_B y_j - c_j$.

Examine the sign of $z_j - c_j$

If all the $z_j - c_j \geq 0$; than initial basic feasible solution x_B is an optimal basic feasible solution; otherwise if any one <0, we have to go to next step.

Now the initial simplex table becomes:



		c =	4	5	9	0	0	
c_B	y_B	x_B	y_1	y_2	y_3	y_4	y_5	
0	y_4	$x_4 = 16$	1	1	(2)	1	0	$\frac{16}{2} = 8$ ←
0	y_5	$x_5 = 25$	7	5	3	0	1	$\frac{25}{3} = 8\frac{1}{3}$
		$z = 0$	-4	-5	-9	0	0	$z_j - c_j$

Step-6: If there are more than one –ve in $z_j - c_j$, choose the most –ve of them, let $Z_r - C_r$. If all $y_{ir} \leq 0$ (i = 1,2,…….m) then there is an unbounded solution to the given problem. If at least one $y_{ir} > 0$, then corresponding vector y_r enters the basis.

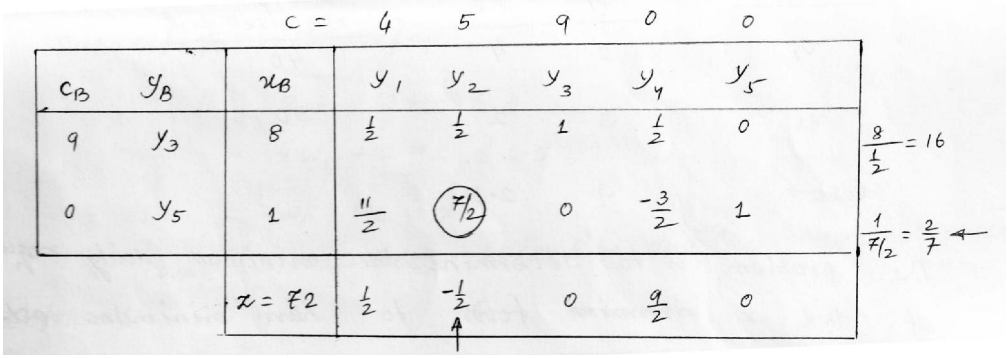
Step- 7: Compute the ratio $\frac{x_{Bi}}{y_{ir}}$, $y_{ir} > 0$, and choose the minimum of them. let the minimum be $\frac{x_{Bk}}{y_{kr}}$, then the vector y_k leaves the basis y_B. The common element is called pivotal element.

Step-8: Convert the pivotal element/leading element and all other elements in the column to zeros.

Step-9: Repeat the computational procedure until either an optimum solution is obtained or there is an indication of unbounded solution.

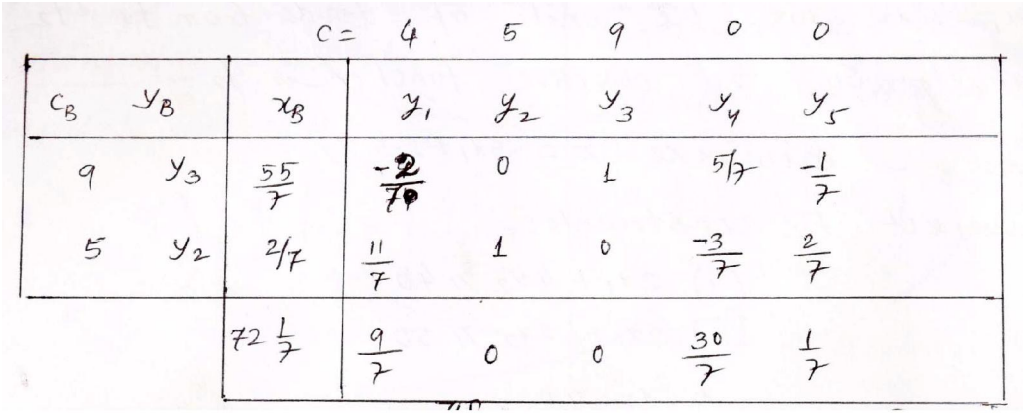
Following the procedure of simplex algorithm, the next simplex table becomes:

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			c = 4	5	9	0	0	
c_B	y_B	x_B	y_1	y_2	y_3	y_4	y_5	
9	y_3	8	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$\frac{8}{\frac{1}{2}} = 16$
0	y_5	1	$\frac{11}{2}$	$\frac{7}{2}$	0	$-\frac{3}{2}$	1	$\frac{1}{\frac{7}{2}} = \frac{2}{7}$ ←
		$z = 72$	$\frac{1}{2}$	$-\frac{1}{2}$ ↑	0	$\frac{9}{2}$	0	

The next simplex table will be :-



			c = 4	5	9	0	0
c_B	y_B	x_B	y_1	y_2	y_3	y_4	y_5
9	y_3	$\frac{55}{7}$	$\frac{-2}{7}$	0	1	$5/7$	$-\frac{1}{7}$
5	y_2	$2/7$	$\frac{11}{7}$	1	0	$-\frac{3}{7}$	$\frac{2}{7}$
		$72\frac{1}{7}$	$\frac{9}{7}$	0	0	$\frac{30}{7}$	$\frac{1}{7}$

Since all zj – cj are ≥ 0 so no repeat procedure is required and the maximum value of z = $72\frac{1}{7}$ ans

© **Concept of duality in simplex methd**

The original LPP is called → primal problem

This original LPP can also be expressed in different form – called → Dual problem.

There is an unique dual problem associated with the primal problem and vice versa.

The following example will clearly explain the duality of original.

Ex: The amount of vitamins (v_1 & v_2) present I 2 different food (f_1 & f_2), cost and daily requirement are presented in the following table.

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Vitamin	Food		Minimum daily requirement
	F1	F2	
V1	2	4	40
V2	3	2	50
Cost	3	2.5	

The problem is to determine the minimum daily requirement of the 2 foods to have minimum cost.

Solution:

In order to formulate the problem mathematically, let x_1 & x_2 are the units of foods from f_1 & f_2 respectively. So the objective function is to.

Minimize $z = 3x_1+2.5x_2$

Subject to constraints:

(i) $2x_1 +4x_2 \geq 40$

(ii) $3x_1 +2x_2 \geq 5$.

Formation of dual problem from this primal problem.

Suppose there is a wholesale dealer who sales 2 vitamins v_1 & v_2 along with other commodities. The dealer knows very well that the foods f_1 & f_2 have market values only because of their vitamin content.

Now the problem of the dealer is to fix maximum per unit selling prices for the 2 vitamins in such a way that the resulting prices of foods f_1 & f_2 does not exceed their existing market price.

Let the dealer decide to fix up 2 prices at w_1 & w_2 per unit respectively. Then the dealer problem can be stated mathematically.

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Maximize $z = 40w_1 + 50w_2$

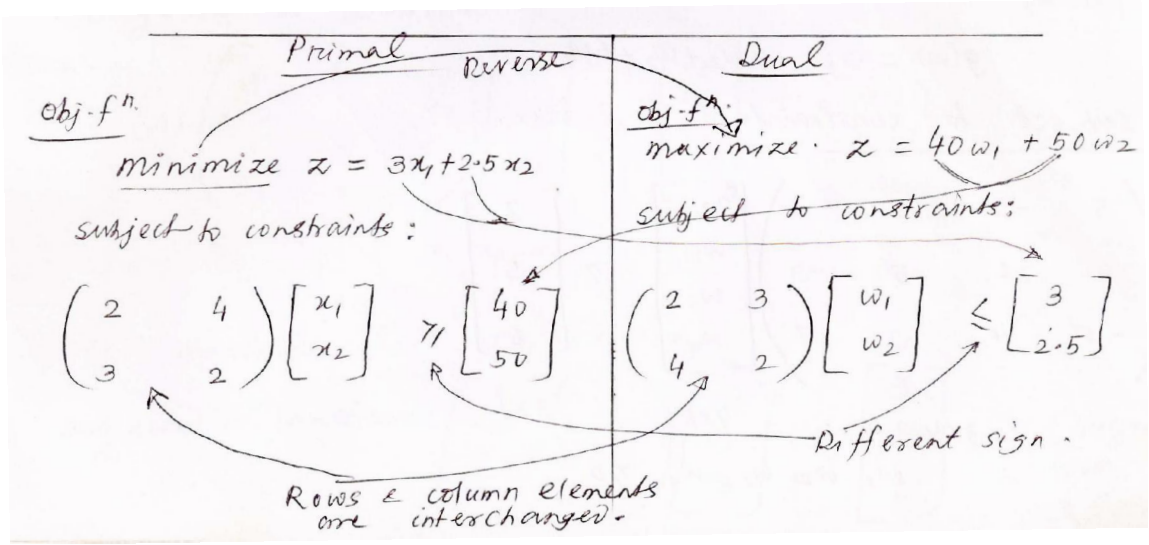
Subject to constraints:

$2w_1 + 3w_2 \leq 3$

$4w_1 + 2w_2 \leq 2.5$

$w_1\ w_2 \geq 0$

A comparison between primal & dual problem is presented as follows:



Ex: Obtain the dual problem of the following LPP.

Maximize f(x) $2x_1+5x_2+6x_3$

Subject to constraints

$5x_1+6x_2-x_3 \leq 3$

$-2x_1+x_2+4x_3 \leq 4$

$X_1-5x_2+3x_3 \leq 1$

$-3x_1-3x_2+7x_3 \leq 6$

$x_1\ x_2\ x_3 \geq 0.$

Also verify that dual of the dual problem is the primal problem.

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Solution: The primal can be restated as : Determine x_1 x_2 x_3 so as to maximize :- $f(x) = 2x_1+5x_2+bx_3$.

Subjected to constraints:

$$\cdot\begin{pmatrix} 5 & 6 & -1 \\ -2 & 1 & 4 \\ 1 & -5 & 3 \\ -3 & -3 & 7 \end{pmatrix}\begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} \leq \begin{bmatrix} 3 \\ 4 \\ 1 \\ 6 \end{bmatrix}$$

4*3 3*1 4*1

Thus if $w = (w_1,w_2,w_3,w_4)$ are the dual variable then the problem is to determine w so, as to minimize.

$d(w) = 3w_1+4w_2+w_3+6w_4$

subject to constraints:

$$\cdot\begin{pmatrix} 5 & -2 & 1 & -3 \\ 6 & 1 & -5 & -3 \\ -1 & 4 & 3 & 7 \end{pmatrix}\begin{bmatrix} w1 \\ w2 \\ w3 \\ w4 \end{bmatrix} \geq \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$$

3*4 4*1 3*1

$w1\ \ w2\ \ w3\ \ w4 \geq 0$

i.e. minimize

$g(w) = 3w_1+4w_2+w_3+6w_4$

subjected to constraints:

$5w_1-2w_2+w_3 - 3w_4 \geq 2$

$6w_1+w_2-5w_3-3w_4 \geq 5$

$-w_1+4w_2+3w_3+7w_4 \geq 6$

$w_1,w_2,w_3,w_4 \geq 0.$

Now, we can restate this dual problem as:

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Maximize h(w) = $-3w_1 -4w_2-w_3-6w_4$

Subjected to constraints

$$(-1)\begin{pmatrix} 5 & -2 & 1 & -3 \\ 6 & 1 & -5 & -3 \\ -1 & 4 & 3 & 7 \end{pmatrix}\begin{bmatrix} w1 \\ w2 \\ w3 \\ w4 \end{bmatrix} \leq \begin{bmatrix} -2 \\ -5 \\ -6 \end{bmatrix}$$

$w_1,w_2,w_3,w_4 \geq 0$

We can write dual of this dual problem as

To minimize

$h(x) = -2x_1-5x_2-6x_3$

subjected to constraints:

$$(-1)\begin{pmatrix} 5 & 6 & -1 \\ -2 & 1 & 4 \\ 1 & -5 & 3 \\ -3 & -3 & 7 \end{pmatrix}\begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} \geq \begin{bmatrix} -3 \\ -4 \\ -1 \\ -6 \end{bmatrix}$$

$x1,x2,x3 \geq 0$

i.e, maximize f(x) = 2x1+5x2+6x3

subjected to constraints:

$$\cdot\begin{pmatrix} 5 & 6 & 1 \\ -2 & 1 & 4 \\ 1 & -5 & 3 \\ -3 & -3 & 7 \end{pmatrix}\begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} \leq \begin{bmatrix} 3 \\ 4 \\ 1 \\ 6 \end{bmatrix}$$ ←original form proceed

© **Dual Simplex Algorithm**

The various steps of dual simplex algorithm in presented in the following.

Step-1: Convert the minimization L.P.P. into maximization if it is in the minimization form.

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Converts into ≥ type inequations representing the constraints into ≤ type by multiplying with (-1).

Step-2: Introduce slack variables in the constraints and obtain an initial basic solution. Put this solution in the starting dual simplex table.

Step-3: Test the nature of $z_j - c_j$ in the starting simplex table.

(a) If all $z_j - c_j$ and x_{Bi} are non-negative for all i & j then an optimum basic feasible solution has been obtained.
(b) If all $z_j - c_j$ are non- negative and at least one basic variable say x_{Bi} is negative, then go to step 4.
(c) If at least one z_j-c_j is negative, the method is not applicable to the given problem.

Step-4: Select the most negative x_{Bis}. The corresponding basic vector then leaves the basis set y_B. Let x_{Bk} be the most negative basic variable so that y_k leaves y_B.

Step-5: Test the nature of y_{ki}, j = 1,2,…….n.

(a) If all y_{ki} are non-negative, there does not exist any easible solution to the given problem.
(b) If at least one y_{ki} is negative, compute the replacement ratios $\{\frac{zj-cj}{yki}$, $y_{ki} < 0\}$ j = 1,2,…….n. and choose the maximum of these. The corresponding column vectors, say y_r (corresponding to $\frac{zr-cr}{ykr}$), then enters the basic set y_B.

Step-6: Test the new iterated dual simplex table for optimality.

Repeat the procedure until either an optimum feasible solution has been obtained (in finite number of steps) or there is an indication of the non-existence of a feasible solution.

Sample problem

Use dual simplex method to solve the LPP

Minimize $z = x_1+2x_2+3x_3$

Subject to constraints:

$X_1-x_2+x_3 \geq 4$

$x_1+x_2+2x_3 \leq 8$

$x_2 - x_3 \geq 2$

$x_1, x_2, x_3 \geq 0$

solution:

the objective function should be maximize z = -x1-2x2-3x3

by introducing slack variables x4, x5, x6 ≥0 in the constraints (after converting into ≤ condition) the equations for constraints would be

$-x_1+x_2-x_3+x_4 = -4$

$X_1+x_2+2x_3+x_5 = 8$

$0\times x_1-x_2+x_3+x_6 = -2$

$$:-\begin{pmatrix} -1 & +1 & -1 & +1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 0 & -1 & +1 & 0 & 0 & +1 \end{pmatrix}\begin{bmatrix} x1 \\ x2 \\ x3 \\ x4 \\ x5 \\ x6 \end{bmatrix} = \begin{bmatrix} -4 \\ 8 \\ -2 \end{bmatrix}$$

Considering x1,x2&x3 = 0 the initial basic solution would be obtained from.

$$.\begin{pmatrix} +1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & +1 \end{pmatrix}\begin{bmatrix} x4 \\ x5 \\ x6 \end{bmatrix} = \begin{bmatrix} -4 \\ 8 \\ -2 \end{bmatrix}$$

i.e.,

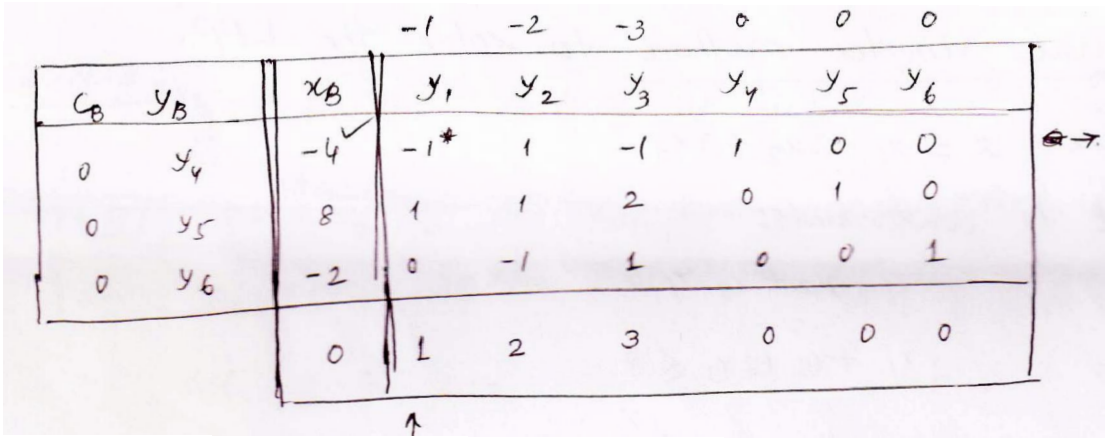
$x_4 = -4,$

$x_5 = 8$

$x_6 = -2$

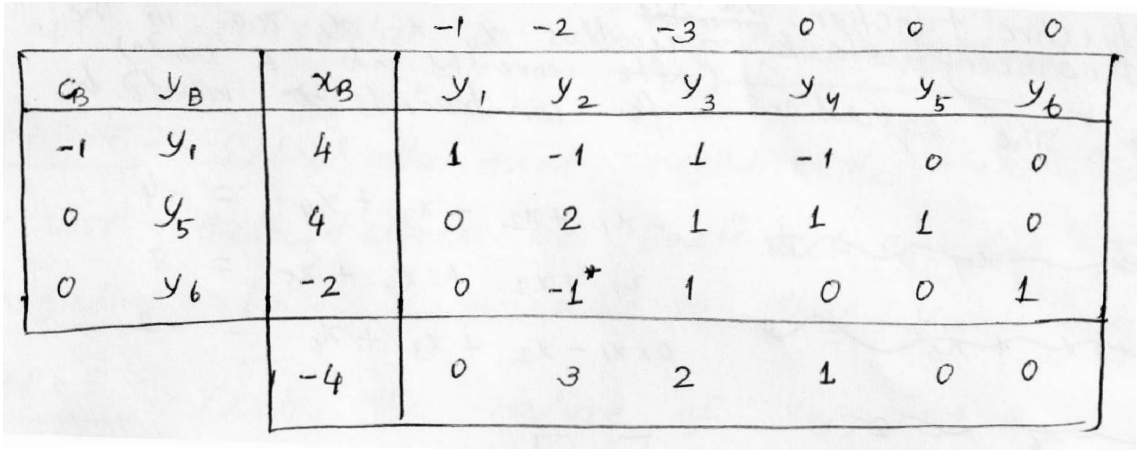
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now the starting dual simplex table:-



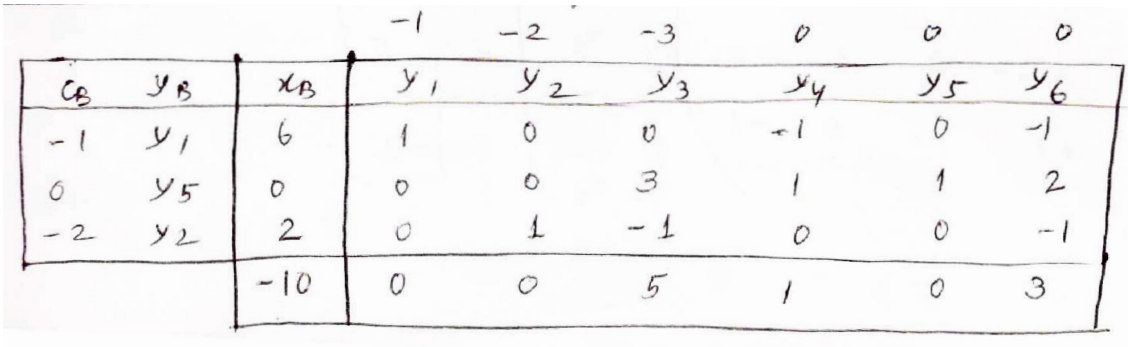
			-1	-2	-3	0	0	0
C_B	y_B	x_B	y_1	y_2	y_3	y_4	y_5	y_6
0	y_4	-4	-1*	1	-1	1	0	0
0	y_5	8	1	1	2	0	1	0
0	y_6	-2	0	-1	1	0	0	1
		0	1	2	3	0	0	0

First iteration: y_4 is dropped and y_1 is introduced. Hence the resulting dual simplex table becomes:



			-1	-2	-3	0	0	0
C_B	y_B	x_B	y_1	y_2	y_3	y_4	y_5	y_6
-1	y_1	4	1	-1	1	-1	0	0
0	y_5	4	0	2	1	1	1	0
0	y_6	-2	0	-1*	1	0	0	1
		-4	0	3	2	1	0	0

Second iteration: y_6 is dropped and y_2 is introduced and the resulting table:-



			-1	-2	-3	0	0	0
C_B	y_B	x_B	y_1	y_2	y_3	y_4	y_5	y_6
-1	y_1	6	1	0	0	-1	0	-1
0	y_5	0	0	0	3	1	1	2
-2	y_2	2	0	1	-1	0	0	-1
		-10	0	0	5	1	0	3

Now since all x_j-$c_j \geq 0$ and also all $x_{Bi} \geq 0$ an optimum basic feasible solution has been reached.

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Hence the optimum basic feasible solution to the given LPP is $x_1=6$, $x_2=2$, $x_3=0$ and minimum z = -(-10) = 10 ans

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